

Laboratory Guide 5: Quantum Measurements

Superposition, State Collapse, Spin & Polarization on the Bloch Sphere

Course: Quantum Computing Foundation

1 Introduction and Objectives

This laboratory offers an **experimental verification and visualization** of the fundamental quantum concepts presented in the *Fundamental Concepts of Quantum Technologies* course. You will explore **superposition**, **quantum measurement**, **photon polarization** and **electron spin**, all represented on the Bloch sphere.

Through guided simulations you will analyze how measurement affects quantum states, how probabilistic outcomes emerge, and how spin and polarization can encode and carry quantum information. The interactive Bloch sphere on the lab's page lets you build intuition for the geometry of quantum states and for the meaning of measurement in different bases.

By the end of the lab you will be able to connect abstract quantum principles to measurable phenomena, describe how measurement defines information in quantum systems, and interpret visual representations of qubit states in terms of real physical quantities.

Prerequisite:

- *Fundamental Concepts of Quantum Technologies* (lecture series).
- Background, when you need it, in the theory pills **QC03** — *The qubit and the Bloch sphere* (qubits, amplitudes, probabilities) and **QC04** — *Single-qubit gates* (the X , Z and H gates).
- Helpful, not required: Labs 1–2 (the Qiskit labs). Nothing in the activities depends on them.

If You Took Labs 1–2 (Optional)

- **The Bloch sphere is not new.** In Lab 1 you used IBM Composer to watch the state vector move as you applied X , H and Z gates. Lab 5 asks a different question: not “how do gates move the state?” but “what happens when we observe it?”
- **The red dot of Lab 1, Ex. 3.** Applying Z after H changed the phase but left the 50/50 probabilities unchanged, because the measurement was in the Z -basis, which cannot see that phase. Activity 2 shows how changing the basis reveals it.
- **Interference vs. measurement (Lab 1, Ex. 5).** The H – Z – H sequence gave $|1\rangle$ with certainty. Activity 5 shows that a measurement in the middle destroys that kind of certainty.
- **Measurement on entangled systems (Lab 2).** Measuring one qubit of a Bell state determined the other: the same state collapse studied here, applied to two qubits at once.

Mode: Interactive Bloch simulator + online simulators **Credits:** 1 ECTS

Learning Objectives:

1. **Connect** abstract quantum principles to measurable physical phenomena.
2. **Describe** how measurement defines and changes information in quantum systems.
3. **Interpret** visual representations of qubit states on the Bloch sphere using the spherical coordinates (θ, φ) .
4. **Distinguish** two operationally different measurement procedures (direct projection vs. rotate-then- Z) and understand when each applies.
5. **Develop** intuition about the probabilistic nature of quantum measurement and the Born rule $P(m) = |\langle m|\psi\rangle|^2$.

Simulator

The lab's interactive page ([lab5_quantum_measurements.html](https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab5/lab5_quantum_measurements.ipynb)) contains a live **Bloch sphere simulator**: six quick-state buttons ($|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle$), sliders for the polar angle θ and the azimuthal angle φ , a choice of measurement basis (Z, X or Y) with the two outcome probabilities shown as bars, a **Measure!** button and an outcome history. The simulator measures by *Procedure 1* (Section 2.4): after a measurement the vector jumps to the basis state that was found. Activities 1, 2, 4 and 5 run on it; Activity 3 uses an external optics simulator. The companion Qiskit notebook (no Qiskit or programming experience is assumed: it opens with a one-screen reading guide):

https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab5/lab5_quantum_measurements.ipynb

2 Theoretical Background

2.1 The Measurement Postulate (Born Rule)

For a qubit in state $|\psi\rangle$ measured in the basis $\{|m\rangle\}$, the probability of outcome m follows the Born rule:

$$P(m | \psi) = |\langle m|\psi\rangle|^2. \quad (1)$$

A *basis* is the pair of perfectly distinguishable states a measurement question can answer, such as $\{|0\rangle, |1\rangle\}$. Here $\langle m|\psi\rangle$ is the *amplitude* of $|m\rangle$ in $|\psi\rangle$. If $|\psi\rangle = a|0\rangle + b|1\rangle$ and $|m\rangle = c|0\rangle + d|1\rangle$, then $\langle m|\psi\rangle = c^*a + d^*b$, where c^* is the complex conjugate of c (change i into $-i$). Example: for $|m\rangle = |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $\langle +|\psi\rangle = (a + b)/\sqrt{2}$. The post-measurement state depends on which procedure is used to perform the measurement (Section 2.4).

2.2 Bloch Sphere Parametrization

Any single-qubit pure state can be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle, \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi). \quad (2)$$

The two amplitudes are $\cos(\theta/2)$ and $e^{i\varphi} \sin(\theta/2)$, so a measurement in the Z -basis gives $P(0) = \cos^2(\theta/2)$ and $P(1) = \sin^2(\theta/2)$: the polar angle θ alone fixes them. The factor $e^{i\varphi}$ is a *relative phase*: invisible to a Z -measurement, visible in the X - and Y -bases (Activity 2). The half angle $\theta/2$ is what makes opposite points of the sphere ($\theta = 0$ and $\theta = \pi$) perfectly distinguishable states. (The theory pill QC03 — *The qubit and the Bloch sphere* introduces this picture.)

state	written out	θ	φ	position
$ 0\rangle$	$ 0\rangle$	0	—	North pole
$ 1\rangle$	$ 1\rangle$	π	—	South pole
$ +\rangle$	$(0\rangle + 1\rangle)/\sqrt{2}$	$\pi/2$	0	$+X$
$ -\rangle$	$(0\rangle - 1\rangle)/\sqrt{2}$	$\pi/2$	π	$-X$
$ +i\rangle$	$(0\rangle + i 1\rangle)/\sqrt{2}$	$\pi/2$	$\pi/2$	$+Y$
$ -i\rangle$	$(0\rangle - i 1\rangle)/\sqrt{2}$	$\pi/2$	$3\pi/2$	$-Y$

Table 1: The six cardinal states of the Bloch sphere.

2.3 Pauli Operators and Eigenstates

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3)$$

Measuring in the Z -basis gives the outcomes $|0\rangle$ or $|1\rangle$; in the X -basis, $|+\rangle$ or $|-\rangle$; in the Y -basis, $|+i\rangle$ or $|-i\rangle$. (X and Z are the gates of the theory pill QC04 — *Single-qubit gates*; here they label the three measurement questions.) The states that give a *certain* answer to a question are called its *eigenstates*. No state gives a certain answer to both the Z and the X question: they are **incompatible**, so sequential measurements in these bases destroy prior information. (Optional, for readers who know matrix products: $[X, Z] = XZ - ZX = -2iY \neq 0$.)

2.4 Two Measurement Procedures

Procedure 1 vs Procedure 2

Same probabilities, different post-states. Throughout this lab, distinguish carefully between two operationally different ways of measuring in a non- Z basis B :

- **Procedure 1 — direct projection.** Project $|\psi\rangle$ directly onto the eigenstates of B . For an outcome $m \in B$, the post-measurement state is $|m\rangle$, an eigenstate of B .
- **Procedure 2 — rotate, then Z -measure.** Apply a reversible gate (a *unitary*) U_B that turns the two states of B into $|0\rangle$ and $|1\rangle$ (the *computational* basis), then perform a Z -measurement. Outcomes are 0 or 1, and the post-measurement state is $|0\rangle$ or $|1\rangle$ — a Z -eigenstate, *not* a B -eigenstate.

The canonical rotations are

$$U_X = H, \quad U_Y = H \cdot S^\dagger, \quad U_Z = I.$$

Here H is the Hadamard gate of the theory pill QC04 — *Single-qubit gates* ($H|+\rangle = |0\rangle$, $H|-\rangle = |1\rangle$); $S^\dagger$ multiplies the amplitude of $|1\rangle$ by $-i$, which turns $|\pm i\rangle$ into $|\pm\rangle$; I does nothing.

The two procedures give the **same outcome probabilities** (Born rule), but the post-measurement states differ. Gate-based quantum processors (such as IBM's) can only execute Procedure 2. Some physical devices measure directly in their own basis and so realize Procedure 1: a polarizer (Activity 3), a Stern–Gerlach magnet (Activity 4), and the simulator on the lab's page.

3 Activities

3.1 Activity 1: State Preparation & Bloch Sphere

Student Activity

Goal. Build visual intuition for qubit states by placing them on the Bloch sphere and understanding the parametrization (θ, φ) of equation (2) (the six cardinal states are in Table 1).

Instructions (the simulator on the lab's page):

1. Click each quick-state button and observe where the vector lands.
2. Drag the θ slider from 0° to 180° and watch the vector travel from the north pole to the south pole.
3. Fix $\theta = 90^\circ$ and drag the φ slider from 0° to 360° . The vector traces the equator, passing through $|+\rangle$, $|+i\rangle$, $|-\rangle$, $|-i\rangle$ and back.
4. Note that changing φ alone does not change the Z -basis probabilities. φ is a **relative phase**: it changes the state ($\varphi = \pi$ turns $|+\rangle$ into $|-\rangle$) but is invisible to a measurement in the Z -basis — Activity 2 makes it visible in the X -basis.

Reflection. Why do $|+\rangle$ and $|-\rangle$ both have $\theta = \pi/2$ but lie on opposite sides of the equator? What does φ physically represent?

3.2 Activity 2: Measurement in Different Bases

Student Activity

Goal. Measure the same state in the Z , X and Y bases and observe how the measurement basis determines the outcome probabilities.

Theory — the Born rule in practice. Prepare $|\psi\rangle = |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Measuring in the Z -basis: $P(0) = P(1) = 50\%$. Measuring in the X -basis: $P(+)=100\%$, $P(-)=0\%$, because $\langle +|+\rangle = 1$ and $\langle -|+\rangle = 0$.

Instructions (the simulator on the lab's page):

1. Click $|+\rangle$. The probability bars show 50% / 50% in the Z -basis — verify this.
2. Switch to the X -basis. The bars now show 100% / 0%. Click **Measure!** several times — always the same outcome. Why?
3. Switch to the Y -basis. The bars return to 50% / 50%. Click **Measure!** once and note the outcome. Click it again: the same outcome repeats, because the first measurement already left the qubit in that Y state. To see the 50/50 randomness, click $|+\rangle$ again before *every* measurement (this is called re-preparing the state).
4. Now try $|-\rangle$: same θ , different φ . In the Z -basis: still 50/50. In the X -basis: the *other* eigenstate becomes certain. In the Y -basis: 50/50 again.
5. For each trial, record: state $\rightarrow$ basis $\rightarrow$ probability bars $\rightarrow$ observed outcome. Watch how the vector snaps to the collapsed position after each measurement.

Reflection. Why does measuring $|+\rangle$ in the X -basis always give the same result? How does this change if you prepare $|-\rangle$ instead?

3.3 Activity 3: Photon Polarization

Student Activity

Goal. Map photon polarization to qubit states, and observe how a polarizer acts as a measurement device and causes state collapse.

Theory — polarization as a qubit. Horizontal polarization $\leftrightarrow |0\rangle$, vertical polarization $\leftrightarrow |1\rangle$, diagonal (45°) polarization $\leftrightarrow |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Linear polarization at angle α (from the horizontal) is $|\alpha\rangle = \cos\alpha|0\rangle + \sin\alpha|1\rangle$. On the Bloch sphere it sits at $\theta = 2\alpha$, $\varphi = 0$ for $0^\circ \leq \alpha \leq 90^\circ$ (for $90^\circ < \alpha < 180^\circ$ at $\theta = 360^\circ - 2\alpha$, $\varphi = 180^\circ$: the anti-diagonal 135° is $|-\rangle$ up to a sign, at $-X$): turning the polarization by 90° (horizontal to vertical) moves the point by 180° (north to south pole).

A polarizer oriented at angle α measures in the basis $\{|\alpha\rangle, |\alpha + 90^\circ\rangle\}$ by *Procedure 1*: a photon that passes is left polarized at α ; one that does not pass is absorbed. A photon with initial polarization angle α_0 passes with probability

$$P(\text{pass}) = \cos^2(\alpha - \alpha_0) \quad (4)$$

— this is Malus's law. For a beam of light it gives the transmitted intensity; for a single photon it is the Born rule: each photon either passes or is absorbed.

Instructions (Quantum Flytrap Virtual Lab, <https://lab.quantumflytrap.com/>):

1. Open Quantum Flytrap and select a single-photon polarization experiment.

2. Prepare a **horizontally polarized** photon and pass it through a **vertical** polarizer. Note: 0% transmission.
3. Prepare a **diagonally polarized** photon (45°) and measure it with a vertical polarizer. Note: 50% transmission.
4. After passing a 45° photon through a vertical polarizer (which leaves it in $|1\rangle$ when it passes), measure again with a horizontal polarizer. Note: 0% transmission. The state has collapsed to $|1\rangle$, which is orthogonal to horizontal.
5. Record each measurement and explain what state the photon is in before and after each polarizer.

Reflection. Why does inserting a polarizer at 45° between two crossed polarizers (horizontal, then vertical) let some light through, even though the crossed polarizers alone block 100% of the light? (Hint: state collapse resets the photon's information.)

3.4 Activity 4: Electron Spin & Stern–Gerlach

Student Activity

Goal. Observe spin quantization and understand how spin-up and spin-down states along different axes map to qubit states and Bloch sphere positions.

Theory — spin and the Bloch sphere. An electron's spin is a two-level quantum system — exactly like a qubit. (Stern and Gerlach's 1922 experiment used silver atoms, whose spin is that of their single outer electron; that is why the instructions below speak of atoms.) Spin-up along Z corresponds to $|0\rangle$ (north pole), spin-down to $|1\rangle$ (south pole). Spin-up along X corresponds to $|+\rangle$ (equator, $+X$); spin-up along Y to $|+i\rangle$ ($+Y$). A Stern–Gerlach apparatus oriented along an axis splits a beam into two outputs (spin-up and spin-down for that axis), each with the probability given by the Born rule (1). Each apparatus implements *Procedure 1*: the post-measurement state is the spin state of the chosen axis that was found.

Instructions (the simulator on the lab's page). A spin behaves exactly like the qubit on the Bloch simulator: a Z -oriented apparatus is a measurement in the Z -basis, an X -oriented one a measurement in the X -basis, and clicking $|0\rangle$ sends a fresh atom from the spin-up output of a first Z apparatus. If your browser can still run it, the PhET Stern–Gerlach simulation (<https://phet.colorado.edu/en/simulations/stern-gerlach>; a legacy Flash simulation) shows the same experiment with beams.

1. Click $|0\rangle$ and measure in the **Z -basis**, several times (click $|0\rangle$ before each): always spin-up. A second Z apparatus confirms the first.
2. Click $|0\rangle$ and measure in the **X -basis** instead, ten times (click $|0\rangle$ before each): about half $|+\rangle$, half $|-\rangle$. Why? Because $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ (check: $|+\rangle + |-\rangle = \frac{2}{\sqrt{2}}|0\rangle = \sqrt{2}|0\rangle$): each of the two terms has amplitude $1/\sqrt{2}$, so each outcome has probability $\frac{1}{2}$.
3. Now take an atom that came out $|+\rangle$ and measure it in the **Z -basis** (a third apparatus). Repeat the whole sequence a few times (click $|0\rangle$, measure in X until the outcome is $|+\rangle$, then measure in Z): $|0\rangle$ and $|1\rangle$ come out about 50/50 — the X measurement erased the Z information.

Reflection. How does this sequential Stern–Gerlach experiment show that the spin- z and spin- x questions are incompatible? What would you expect if a spin could have a definite z value and a definite x value at the same time?

3.5 Activity 5: Sequential Measurements & State Collapse

Student Activity

Goal. Demonstrate that measuring in Z , then X , then Z again produces random outcomes in the final Z measurement, even when the first Z gave a definite result.

Theory — incompatible questions. The Z question (“ $|0\rangle$ or $|1\rangle$?”) and the X question (“ $|+\rangle$ or $|-\rangle$?”) have no common eigenstate: $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, so a state that answers one question for certain answers the other with 50/50. (Optional, for readers who know matrix products: $XZ - ZX = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} = -2iY \neq 0$.)

Which procedure? The simulator on the lab’s page measures by **Procedure 1**: after the X -measurement the qubit is left in $|+\rangle$ or $|-\rangle$. A gate-based processor would measure X by **Procedure 2** — apply $U_X = H$, then Z -measure — and leave $|0\rangle$ or $|1\rangle$ instead. The final Z statistics are the same either way (50/50 across many runs); what differs is the state in between. Self-Assessment question 3 asks about the Procedure 2 version.

Instructions (the simulator on the lab’s page):

1. Click $|0\rangle$, select the Z -basis and click **Measure!**: always $|0\rangle$. Post-state: $|0\rangle$.
2. **X-measurement**: switch to the X -basis (the bars show 50% / 50%) and click Measure!. The vector jumps to $|+\rangle$ or $|-\rangle$ on the equator.
3. **Second Z-measurement**: switch back to the Z -basis (the bars show 50% / 50% again) and click Measure!: $|0\rangle$ or $|1\rangle$. The X -measurement erased the Z information.
4. Repeat steps 1–3 ten times (click $|0\rangle$ each time) and count how often the last outcome is $|0\rangle$: about half the time.
5. **Compare to $Z \rightarrow Z \rightarrow Z$** : click $|0\rangle$ and measure in the Z -basis three times in a row. The outcome is always $|0\rangle$: a repeated question does not disturb the answer.

Reflection. The probabilities of the final Z measurement are 50/50 (across many runs) because the X step left the qubit in $|+\rangle$ or $|-\rangle$, each of which gives 0 or 1 with probability 1/2. Can the final Z measurement ever recover the original Z information? Why not? What property of the two questions makes this fundamental?

4 Simulator Tools

Activities 1, 2, 4 and 5 run on the Bloch simulator of the lab's page. The external tools below open in a separate browser tab; no installation is needed.

- **Quantum Flytrap — Virtual Lab** (<https://lab.quantumflytrap.com/>): photon polarization experiments with beam splitters, wave plates and polarizers. Activity 3.
- **QuVis — Bloch sphere & measurement** (<https://www.st-andrews.ac.uk/physics/quvis/>), University of St Andrews. Optional: Activities 1–2 use the simulator on the lab's page.
- **PhET — Stern–Gerlach experiment**, University of Colorado Boulder: <https://phet.colorado.edu/en/simulations/stern-gerlach>. Activity 4, optional (legacy Flash).
- **IBM Quantum Composer** (<https://quantum.ibm.com/composer>). Optional, if you took Lab 1: the Activity 5 sequence as a circuit (Procedure 2).

5 Self-Assessment

Self-Assessment

Question 1 — the Born rule. A qubit is prepared in the state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$. What is the probability of measuring the outcome $|1\rangle$ in the Z -basis?

- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$

Question 2 — state collapse: procedure matters. Starting from $|\psi\rangle = |0\rangle$, you apply H and then perform a Z -basis measurement. The outcome is 0. What is the post-measurement state?

- (a) $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$
(b) $|0\rangle$
(c) $\frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$ (renormalized)

Question 3 — non-commutativity. You prepare $|0\rangle$, measure Z (get 0, post-state $|0\rangle$), then measure X via Procedure 2 (apply H , then measure in the Z -basis), then measure Z a final time. What are the probabilities for the final Z measurement, over many runs (the middle outcome is random)?

- (a) 100% outcome 0 — the initial Z measurement fixed the state.
(b) 50% / 50% — the X measurement randomized the qubit between $|0\rangle$ and $|1\rangle$.
(c) 0% outcome 0 — measuring X always flips the Z information.

6 Resources

- Theory pills QC03 — *The qubit and the Bloch sphere* and QC04 — *Single-qubit gates*. — The background this guide points to.
- Nielsen, M. A. & Chuang, I. L., *Quantum Computation and Quantum Information*, Cambridge University Press, 2010, §2.2.3 (the measurement postulate) and §1.2 (the Bloch sphere).
- Barnett, S. M., *Quantum Information*, Oxford University Press, 2009, §2.4 (qubits) and §4.1 (ideal von Neumann measurements).
- Preskill, J., *Lecture Notes on Quantum Information*, Ch. 2 (foundations).
- Online: Qiskit documentation (<https://qiskit.org/documentation/>) — statevector and measurement simulation; PennyLane QML (<https://pennylane.ai/qml/>) — quantum measurements and observables; QuVis learning modules — Bloch sphere and measurement tutorials.
- Companion Jupyter notebook, in the repository `quantum-labs-2026:lab5/lab5_quantum_measurements.ipynb` (Colab link in the Simulator box of the Introduction).